

Keeping Score on Two Centuries of Technological Unemployment Predictions

The Pilot Phase of the Tech-Panic Tracker

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1 Introduction

Predictions that machines will take human work are almost as old as industrial machinery itself. Framework knitters petitioned against wide frames in the 1780s, Ricardo added his chapter “On Machinery” to the third edition of the *Principles* in 1821 (Ricardo 1821), Keynes named “technological unemployment” for a wide audience in 1930 (Keynes 1930), Herbert Simon expected machines capable of any human work “within twenty years” in 1960 (Simon 1960), and Frey and Osborne put 47 per cent of United States employment in a high-risk category in 2013 (Frey and Osborne 2013). The current wave of generative-artificial-intelligence forecasts is the most recent instalment in a series that runs for more than two centuries and recruits economists, futurists, government agencies, intergovernmental organisations, consultancies, and labour leaders alike. The historical shape of this discourse has been described: Mokyr, Vickers, and Ziebarth (2015) survey two centuries of technological anxiety in the *Journal of Economic Perspectives*, and Woirol (1996) traces the professional debates of the 1930s and 1960s in detail. However, no systematic and open dataset collects these predictions across the full historical span, classifies them on a common scheme, and evaluates them against outcome data over the horizons their authors stated.

The absence is a curious one, because economics does institutionalise forecast evaluation. The International Monetary Fund’s *World Economic Outlook* forecasts have been audited repeatedly (Timmermann 2007), the Federal Reserve Bank of Philadelphia maintains a real-time evaluation programme for its Survey of Professional Forecasters, and Loungani’s assessment of recession forecasting is well known (Loungani 2001). All of this machinery is pointed at short-horizon quantitative macroeconomic series. The predictions that actually shape technology policy are long-horizon structural claims, and those have so far been evaluated only one at a time: Georgieff and Milanez (2021) and Coelli and Borland (2019) on Frey and Osborne, Turner (2008) and Herrington (2021) on *The Limits to Growth*, Keilman (2001) on United Nations population projections. Each of these is a careful retrospective on a single forecast. Nobody maintains a database of long-horizon structural predictions.

The nearest methodological neighbour is Armstrong and Sotala’s collection of roughly 257 artificial-intelligence predictions, assembled for the Machine Intelligence Research Institute and still downloadable (Armstrong, Sotala,

and Ó hÉigeartaigh 2014). That dataset covers artificial-intelligence timelines only, from the 1950s onward, and it does not evaluate predictions against outcomes. It also carries an instructive lesson: the headline finding that expert and non-expert predictions were indistinguishable did not survive re-analysis, because of a documented coding error. Coding quality and audit trails are precisely what a project of this kind must defend, and the verification log described in Section 3 is our answer to that hazard. The Pessimists Archive, meanwhile, is a valuable source of leads and a good deal of the motivation for this project, but it is a curated advocacy collection with an editorial thesis of its own, no fixed corpus, no schema, and no versioning. It should be mined for clippings and never treated as data.

This paper introduces the Tech-Panic Tracker, an open dataset of labour-replacement predictions from approximately 1800 to the present, and reports the results of its pilot phase. The project asks four questions. The first is descriptive: how are these predictions distributed over time, by technology, by author type, by level of aggregation, and by how specific they were? The second is evaluative: of the predictions whose horizons have elapsed, what share were correct, and to what degree on a graded rubric? The third is explanatory: which features of a prediction are associated with its accuracy, among horizon length, technology generality, author type, whether a mechanism was specified, and whether the claim carried a number? The fourth is behavioural: do authors ever return to their own predictions to reaffirm, moderate, quietly extend a deadline, or retract, and does revisiting bear any relation to having been right? The pilot reported here touches all four, but it settles none of them; a pilot of twenty-three purposively selected predictions cannot.

The contribution the full project aims at is a systematic, all-author-type, two-century, evaluated, open dataset. Mokyr, Vickers, and Ziebarth (2015) told the story of technological anxiety; the ambition here is to build the sampling frame and keep score, in roughly the way that Tetlock generalised the checking of individual pundits into a research programme (Tetlock 2005). The retrospective evaluation of a single famous forecast is by now a recognisable genre, and this project attempts to generalise that genre into an instrument.

The remainder of the paper proceeds as follows. Section 2 states what a pilot conducted before preregistration is for, and what it deliberately does not attempt, because the purposive design of the corpus governs how every subsequent number should be read. Section 3 describes the relational data model and the verification pipeline, whose central constraint is that a language model may draft a record but only a human may advance it. Section 4 sets out the two-track search protocol and the era stratification that the search vocabulary requires, and reports the pilot's review of the evidence behind each era boundary, which produced the one substantive finding of this phase. Section 5 presents the evaluation rubric, including the separate treatment of exposure and capability claims that the design turns on. Sections 6 through 8 report what the pilot produced: the corpus and its distributions, six trial evaluations, and a demonstration run of the extraction pipeline on a passage the model had not seen. Section 9 discusses what the pilot established and what it revealed about the cost structure of the full collection. Section 10 states the decisions that must be taken jointly before Registration 1, and Section 11 sets out the roadmap.

2 Goals of the Pilot

A pilot conducted before preregistration exists so that changes to the schema and the rubric are easy and honest. Once the search and extraction protocols are registered, every vocabulary addition becomes a logged amendment and every rubric change after the first verdict requires reporting results under both the old and the new rules. Before registration, the same changes cost nothing but the time to make them. This sequencing is stated openly in the registrations, and reviewers generally regard piloting before registering as good practice rather than as a weakness.

The pilot has four objectives. The first is to establish that the schema can hold real historical material without distortion, which is tested by feeding it some awkward cases rather than easy ones. The second is to run the full pipeline from source text through structured extraction to a graded verdict with cited evidence, so that the feasibility of the whole apparatus is demonstrated. The third is to stress-test the evaluation rubric on a small number of predictions whose horizons have elapsed, in order to find the places where the bands or the decision rules give the wrong answer or no answer at all. The fourth is to measure the costs that the full collection will incur, from per-database query yields to the labour of obtaining archive permalinks.

The corpus assembled for these purposes is illustrative, not really representative, and this governs how every distribution in Section 6 should be read. The twenty-three predictions were selected from the main study / seed list to cover the corners of the schema: all five era strata, all four specificity tiers, all three claim types, both alarms and reassurances, named individuals and institutional authors, and the two author-type values added during the pilot. A representative sample would have concentrated in the modern strata where the sources are dense and the retrieval

is easy, and would have exercised very little of the schema. The proportions reported below are therefore properties of a deliberately awkward selection and carry no information about the underlying discourse.

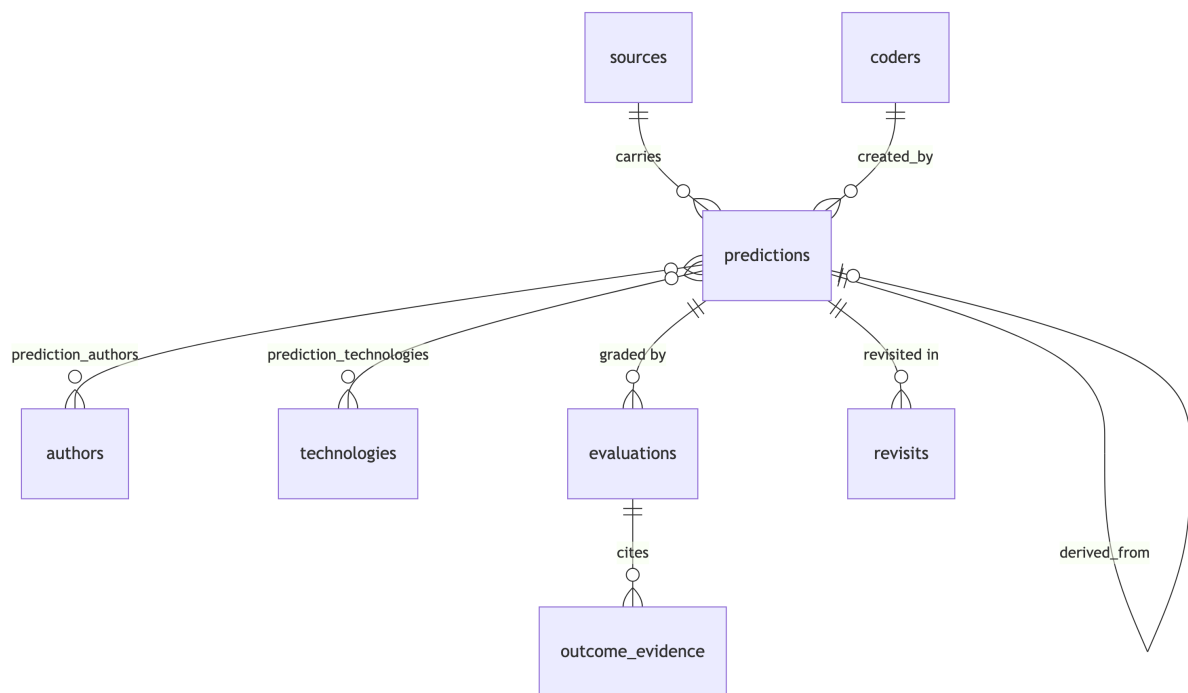
The pilot also doesn't attempt several things that the full project requires. It does not run the systematic database search, though it does measure numbers for the draft queries. It does not compute human-human reliability, because the records have not yet been double-coded. It does not produce registered verdicts, because the rubric that generated the six verdicts in Section 7 has not been frozen. What it produces is a working instrument and a list of decisions.

3 Data Model and Verification Pipeline

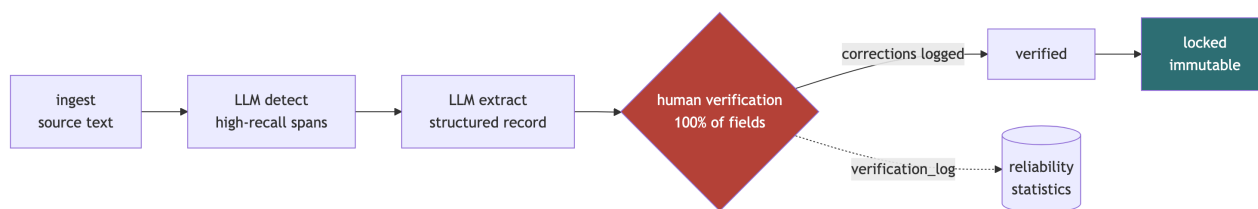
! Discussion of the use of LLMs for extraction

Ben we need to talk about whether we are interested in using LLMs for the first pass in the extraction process.

The dataset comprises nine tables and three logs, all held as plain comma-separated files under version control on GitHub. The core table is `predictions`, and the modelling decision that matters most is that a source and a prediction are distinct entities: one report may carry several distinct claims, and one claim may be reported by dozens of documents. Collapsing the two, which would be the natural simplification, would make it impossible to distinguish a consultancy that made three separate forecasts from a newspaper that quoted one forecast three times. Every prediction record carries a mandatory verbatim quotation and a mandatory locator, so that a hallucinated quotation fails verification by construction.



Two features of the model are worth drawing out, because both answer specific hazards. The self-referential relation at the lower right, in which a prediction may point to another prediction through `derived_from_prediction_id`, records citation lineage; Section 6 shows why it is necessary. The `revisits` table records occasions on which an author returned to a claim, with a stance drawn from a controlled list that includes reaffirmation, moderation, deadline extension, retraction, reversal, and silent contradiction. Recording a revisit separately from the original prediction means that an author's later framing never overwrites what they said at the time.



The pipeline runs in five stages, and the diamond in the diagram is the constitutional constraint on all of them: the model drafts and a human decides. Candidate detection is tuned for high recall, because a false positive costs a coder a few seconds while a false negative is a big problem! Structured extraction fills a schema-constrained object mirroring the predictions table. Nothing held in the interim directory is ever analysed, and every field that a human changes at verification is written to `verification_log` with the model’s value and the human’s value side by side. That log is published with the dataset, and it is what will later permit honest agreement statistics between model and coder rather than an assurance that the coders were careful. Because every analysed record is human-verified, language-model performance is a workload and quality metric rather than a validity threat, and the preregistration states this framing explicitly.

Model and prompt versions are recorded on every record, and the model identifier is the exact dated snapshot returned by the interface rather than a floating alias. An alias that silently rolls forward would mean that the preregistration no longer describes what actually ran, and would break the frozen regression set of documents that is re-run whenever a prompt or model changes.

4 Search Protocol and Era Stratification

The search runs on two tracks. Track A is a set of reproducible query strings executed against registered databases and logged, query by query, with result counts and export counts. Track B is a documented snowball and institutional-publisher sweep, conducted to PRISMA-S conventions, with citation chasing capped at two generations per seed and every hop recorded through a parent-source identifier. The hybrid is a concession to the material rather than a preference: strict PRISMA is not achievable for nineteenth-century newspapers or for consultancy reports that databases index poorly, and a purely systematic search would silently exclude the grey literature where a lot of institutional forecasts live. Both tracks feed the same screening stages, and accuracy is reported separately by retrieval track, because famously wrong predictions are precisely what gets cited and Track B is therefore exposed to salience bias.

Volume management is registered rather than improvised. Any stratum-by-database cell whose yield exceeds the screening cap is screened on a simple random sample drawn within stratum and year by a seeded, scripted procedure, sampling probabilities are stored per source, and the resulting weights propagate into every corpus-level statistic, with unweighted versions reported as robustness. Cells under the cap are just screened exhaustively. The pilot’s function here is to measure the yields that set the cap.

The search vocabulary must change by era, because the words change. A query for “automation” returns nothing useful before the late 1940s, and a query for “labour-saving machinery” returns very little after the First World War. Table 1 gives an abridged version of the era-stratified vocabulary.

Table 1: The era-stratified query vocabulary, abridged. Notes: Each stratum crosses a technology-term block with a displacement-term block; the full registered query strings, database-specific syntax, and date filters are held in the search protocol. Plain hyphens are used in period ranges throughout the tables in this document because the Typst renderer mangles en-dashes inside table cells.

Stratum	Period	Technology terms	Displacement terms
E1	1800-1870	labour-saving machinery, the machinery question, power loom	thrown out of work, superfluous workmen, distress of the operatives
E2	1870-1920	mechanisation, the machine age, automatic machinery	displacement of labour, machinery and unemployment
E3	1920-1955	technological unemployment, automation (from 1947), robot (from 1921)	men replaced by machines, permanent unemployment
E4	1955-1995	automation, cybernation, robotics, microelectronics	jobs destroyed, end of work, silicon collar
E5	1995-	computerisation, artificial intelligence, machine learning, generative AI	jobs at risk, automation of jobs, replace workers

4.1 The Evidence Behind the Era Boundaries

The five era bands are chosen by the authors, and here we provide the evidence and reasoning behind each boundary.

The 1870 boundary should be uncontroversial. Mokyr (1999) dates the Second Industrial Revolution to the years 1870 to 1914, and Gordon’s growth periodisation breaks in the same place (Gordon 2012). One caution is worth recording, because it caught us out during the pilot. Landes’s *The Unbound Prometheus* is cited for this break constantly, but his own chapter seam falls at 1873, he treats it as the end of the first industrial revolution rather than the beginning of a second, and he places the second cluster of innovations in the 1890s while holding the label itself at arm’s length (Landes 1969, 196, 235). He remains a good citation for 1870 as a break, in his preface, as the moment continental Europe caught up with Britain, and a poor one for anything else.

The 1955 boundary is also sensible. Delmar Harder established an Automation Department at Ford in April 1947 (Hounshell 1995, 70), the word appeared in print in *American Machinist* in October 1948, Diebold’s *Automation* followed in 1952 (Diebold 1952), and by October 1955 Congress was running nine days of hearings on the subject. Wright Patman, chairing, observed that the word “does not appear in recent dictionaries although we can scarcely pick up a magazine without finding a reference to it” (United States Congress, Joint Committee on the Economic Report 1955, 51). A word absent from the dictionaries and already generating congressional hearings is what an era boundary ought to look like.

The 1995 boundary is based on Oliner and Sichel’s dating of the productivity acceleration (Oliner and Sichel 2000, 3), with the caution that Gordon’s own growth-accounting break falls at 1994 (Gordon 2016) and he is therefore the wrong authority to cite for it. The 1800 boundary is a convention rather than a break: the machinery debate runs back at least to Steuart (1767), and the protocol should say so rather than allow the table to imply a discontinuity that nobody claims.

Perhaps 1920 should be changed to 1928? The first appearance in print of the phrase “technological unemployment” is an unsigned *New Republic* article of 8 February 1928, reprinted into the *Congressional Record* on 21 February 1928 at Senator Wagner’s request, where it sits in scare quotes: “the problem of ‘technological’ unemployment will be with us permanently” (“The Price of Industrial Progress” 1928). The Hoover Committee was still calling it “what has come to be known as ‘technological’ unemployment” and “this newer problem” in 1929 (Committee on Recent Economic Changes of the President’s Conference on Unemployment 1929, xvii). The peer-reviewed literature dates the famous Debate on Technological Unemployment to the years 1928 to 1933 (Katselidis, Vouldis, and Michaelides 2011). Amy Sue Bix’s history of the American debate opens in 1929 (Bix 2000). The full evidence is recorded in the decision register, and the consequences of moving the boundary are discussed in Section 10.

! To discuss with Ben

I went looking for someone to stand behind 1920 and came back with nothing. Everything I found points at 1928.
While we're in here: do the five eras seem sensible to you?

5 Evaluation Rubric

Verdicts are outcome-based: the question is whether the predicted employment state occurred, in the stated place, by the stated date, regardless of what caused it. The alternative, requiring causal attribution before assigning a verdict, would turn every single verdict into its own research project and would fill the dataset with the terminal code for insufficient evidence. The cost is real, and the design records it rather than engineering around it: an exploratory attribution-confidence field captures the evaluator's judgement of whether the named technology plausibly drove the outcome, and that field never gates a verdict. The limitation is stated openly in the paper rather than buried.

For quantified claims the verdict follows from the ratio of the realised magnitude to the claimed magnitude at the horizon year. Table 2 sets out the bands. They are two-sided and symmetric in log space, because over-realisation is also a failure of magnitude; a prediction that a hundred thousand jobs would be lost is not clearly correct if four hundred thousand were lost.

Table 2: The verdict bands for quantified claims. Notes: The three terminal codes - too early to tell, unfalsifiable, and insufficient evidence - sit outside the ordinal scale and are reported as outcomes rather than dropped. Directional (T3) claims are capped at mostly correct, so that vague predictions are not structurally advantaged over brave quantified ones; rhetorical (T4) claims are unfalsifiable by construction unless a horizon default rescues an operationalisable core. Band edges are stress-tested in the pilot and frozen at Registration 2.

Verdict	Ratio r = realised / claimed at the horizon year
clearly correct	2/3 to 3/2
mostly correct	0.4 to 2/3, or 3/2 to 2.5
mixed	0.1 to 0.4, or 2.5 to 10, or right direction and wrong locus
mostly wrong	below 0.1 or above 10, direction intact
clearly wrong	direction contradicted

Three decision rules do the substantive work, and each answers a foreseeable objection. The first is that exposure claims are graded on their own track. A claim that 47 per cent of jobs are at high risk of computerisation is a claim about susceptibility, not about realised job loss, and the appropriate test is differential: over the horizon, did the occupations the source classed as high-exposure decline, in employment or in growth rate, relative to those it classed as low-exposure? Grading such a claim as though it forecast the disappearance of 47 per cent of jobs scores a claim the authors never made, and it is the single most predictable referee objection to the whole project. Sources that give an aggregate exposure number without publishing an occupation-level ranking cannot be tested this way and are recorded as unfalsifiable on that track rather than graded as displacement forecasts.

The second rule is that capability claims are tested on capability. Simon claimed that machines would be able to do the work; whether firms subsequently hired less people is a different question with a different answer, and pooling the two would confound optimism about technology with optimism about its labour-market consequences. Keeping the tracks separate is what will eventually allow the exploratory question of whether predictions fail at the capability stage or at the link from capability to employment.

The third rule is that every ordinal verdict must cite evidence, and this one is enforced in code rather than in good intentions. The validation pipeline rejects a graded verdict with no outcome-evidence row, and rejects a verdict at either extreme of the scale whose evidence is not official statistics, a census series, or an academic study. Section 7 shows the rule doing its work.

Vague horizons are operationalised through a preregistered lookup table rather than at the evaluator's discretion: "soon" and its relatives default to ten years, "within a generation" to thirty, "in our lifetime" to forty, and "in the long

run” yields no horizon at all unless an explicit magnitude is present, in which case a fifty-year convention applies and the record is flagged. Every defaulted case sets a flag, and the registered robustness suite re-runs the headline analyses with all defaulted horizons excluded.

6 The Pilot Corpus

The pilot corpus contains 23 predictions drawn from the landmark seed list, and all 23 rows pass both the frictionless schema and the six cross-table validation rules. As set out in Section 2, they were selected to cover the corners of the schema rather than to be representative, and the distributions below should be read accordingly. Table 3 summarises where the corpus sits by source type and era stratum, and Table 4 and Table 5 list the records themselves.

Table 3: Source type by era stratum, pilot corpus. Notes: Era strata are E1 (1800-1870), E2 (1870-1920), E3 (1920-1955), E4 (1955-1995), and E5 (1995 to present). Counts are of predictions, not of sources; one source may carry several predictions. Selection is purposive, so the distribution reflects the coverage design described in Section 2 rather than the composition of the discourse.

source type	E1	E2	E3	E4	E5
book	2	1	0	1	1
consultancy_report	0	0	0	0	3
government_report	0	1	0	0	0
igo_report	0	0	0	0	1
journal_article	0	0	0	1	1
magazine_article	0	0	2	0	0
newspaper_article	0	0	0	0	1
pamphlet	1	1	1	0	0
speech_transcript	0	0	0	0	1
testimony	1	0	0	1	0
think_tank_report	0	0	0	0	1
working_paper	0	0	0	0	1

The awkward cases are the point of the exercise. The corpus contains a pseudonymous threatening letter and a Goldman Sachs research note, a maiden speech in the House of Lords and a recording of a conference talk hosted on a video platform, six reassurances against fourteen alarms, all four specificity tiers, all three claim types, and both of the author-type values added to the vocabulary during the pilot. Each of these was entered because it was expected to break something, and the schema revisions that followed are recorded in the decision register.

Table 4: The pilot corpus, 1812-1949. Notes: Author type is the `author_type_at_prediction` field from the prediction-authors junction table, coded as of the prediction date rather than from the author’s later career. Tier abbreviates the specificity tier: T1 carries both an explicit magnitude and an explicit horizon, T2 carries one of the two, T3 is directional with a clear locus but no numbers, and T4 is rhetorical with no operationalisable claim. Posture is the `panic_valence` field. The one-sentence claim summary for each row is held in `predictions.csv` and is omitted here because the column overflows the page.

Year	Who	Author type	Source	Tier	Posture
1812	Ned Ludd	labor organization	pamphlet	T4	alarm
1812	Byron	politician	testimony	T3	alarm
1821	Ricardo	academic	book	T3	neutral
1835	Ure	academic	book	T3	reassures
1886	Wright	government agency	government report	T3	reassures
1889	Wells	academic	book	T2	alarm
1894	Gompers	labor organization	pamphlet	T3	alarm
1930	Keynes	academic	magazine article	T2	neutral
1938	Compton	academic	magazine article	T3	reassures
1949	Wiener	academic	pamphlet	T3	alarm

Table 5: The pilot corpus, 1955-2025. Notes: Columns are defined as in Table 4.

Year	Who	Author type	Source	Tier	Posture
1955	Reuther	labor organization	testimony	T3	alarm
1960	Simon	academic	book	T1	neutral
1983	Leontief	academic	journal article	T3	alarm
1995	Rifkin	futurist	book	T2	alarm
2013	Frey & Osborne	academic	working paper	T2	alarm
2014	Deloitte	consultancy	consultancy report	T1	alarm
2015	Autor	academic	journal article	T3	reassures
2016	OECD / Arntz	academic	igo report	T1	reassures
2016	Hinton	academic	speech transcript	T1	alarm
2017	PwC	consultancy	consultancy report	T1	alarm
2022	ITIF / Atkinson	think tank private	think tank report	T2	reassures
2023	Goldman Sachs	financial institution	consultancy report	T1	alarm
2025	Amodei	industry executive	newspaper article	T1	alarm

An earlier version of these two tables carried the one-sentence claim summary as a sixth column. The Typst renderer laid out the headers and then silently dropped every row, because the cell was too wide to place, and no error was raised at any point in the pipeline. The behaviour is worth recording here, because it is the failure mode that would quietly delete a results table from the finished paper.

6.1 Vocabularies and Waves

The discourse in which these predictions sit has a shape, and it is not a trend. Figure 1 plots three phrases from the Google Books English corpus over the whole period, each indexed to its own maximum.

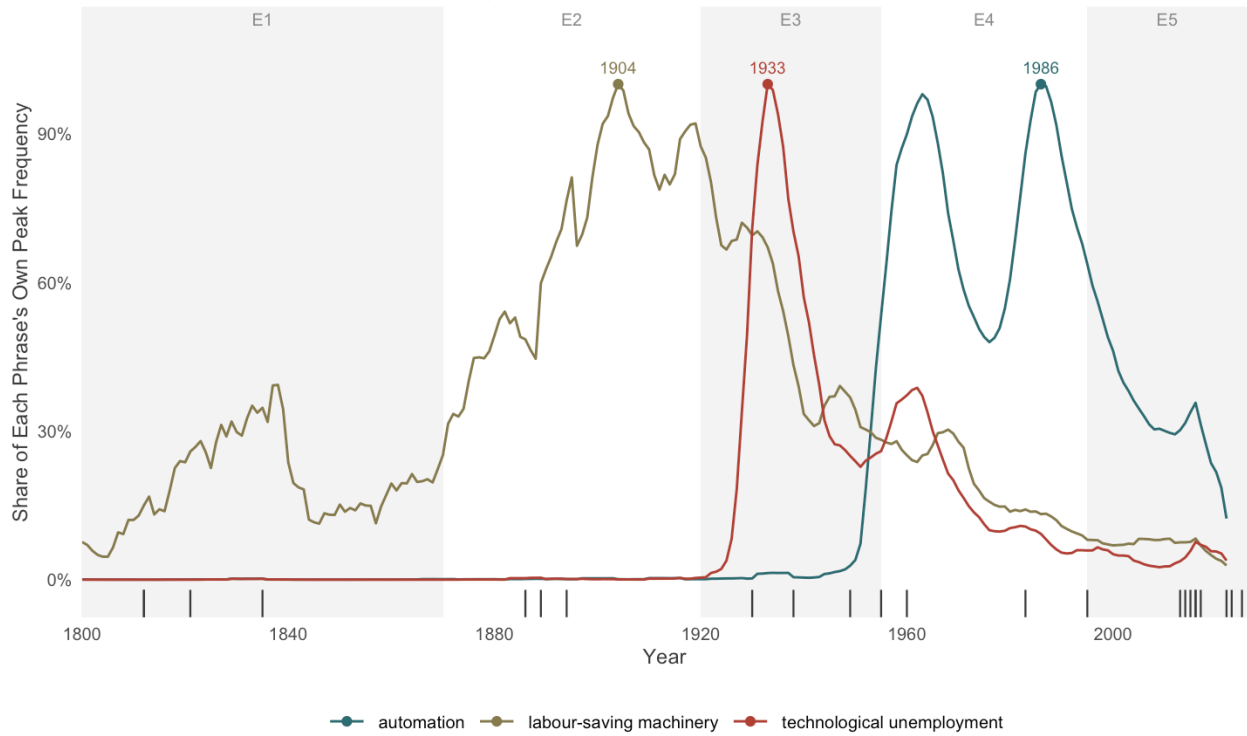


Figure 1: Relative frequency of three displacement vocabularies in the Google Books English corpus, 1800-2022, each series indexed to its own peak. Notes: The vertical axis is each phrase’s frequency as a share of that phrase’s own maximum, so the three series are comparable in shape but not in level; the year of each peak is printed beside the marker. Indexing is necessary because ‘automation’ is roughly twenty-five times more common in print at its peak than ‘technological unemployment’ ever was, and on a common absolute scale the two older phrases are invisible. Alternating background bands are the five era strata, labelled E1 (1800-1870) through E5 (1995 to present). Tick marks along the lower axis mark the prediction dates of the 23 pilot predictions. Sources: Google Books Ngram, English corpus, smoothing parameter 3. Ngram frequency is a proxy for discourse volume and not for the number of predictions made; the registered time-series descriptive in the full paper is a normalised rate of query hits per million digitised pages per archive-year, computed from the registered sources, with raw counts reported alongside.

The figure shows three vocabularies and three waves, each largely spent before the next begins: “labour-saving machinery” peaks in 1904, “technological unemployment” in 1933, and “automation” in 1986. The pattern is what justifies the era-stratified query design of Section 4, since a single fixed vocabulary applied across the whole period would retrieve each wave with quite different efficiency and would confound the shape of the discourse with the shape of the query. The comparison of absolute volume between phrases is lost by indexing, but that comparison is a fact about the English language rather than about panics, and little is given up.

Figure 2 places the twenty-three pilot predictions in time, with rhetorical posture on the vertical axis.

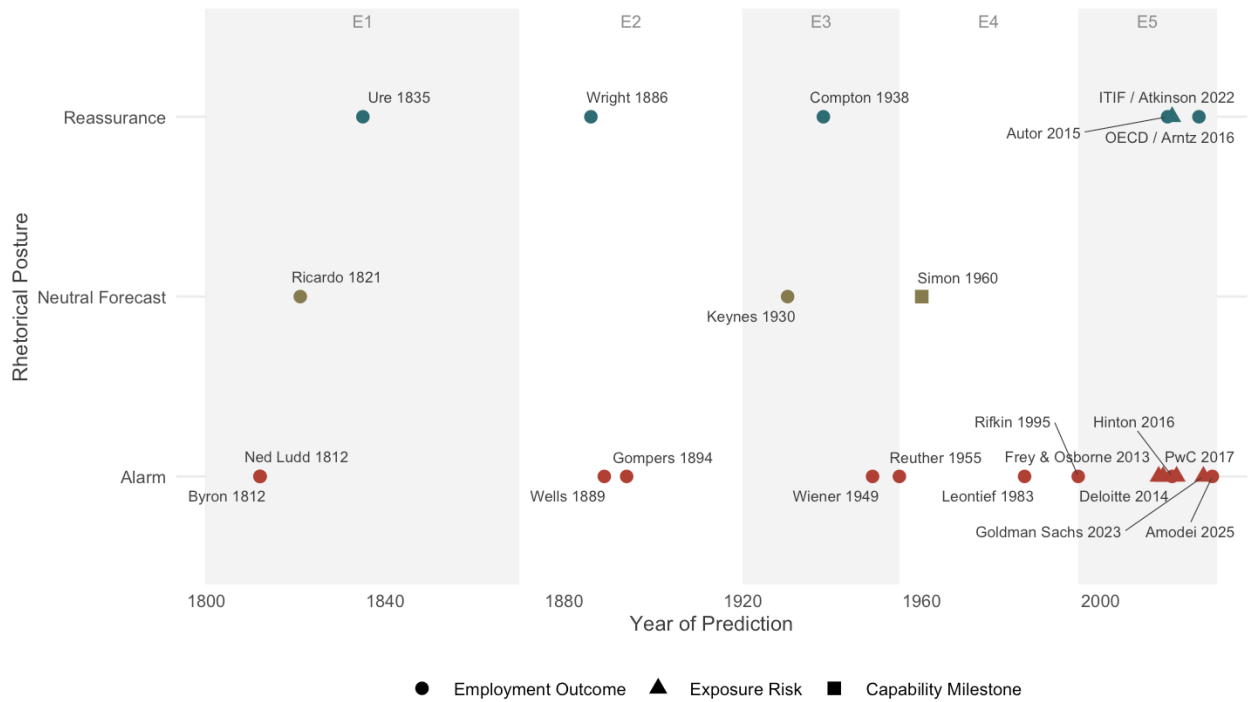


Figure 2: The 23 pilot predictions by year and rhetorical posture, 1812-2025. Notes: Vertical position is the panic_valence field: alarm (a technology will destroy employment), neutral forecast, or reassurance (a technology will not destroy employment, or will create more work than it displaces). Marker shape is the claim type: an employment outcome is a claim that jobs will be lost or created, an exposure risk is a claim that a share of jobs is susceptible to automation without asserting that the loss will be realised, and a capability milestone is a claim that machines will become able to perform some class of work. Labels give the lead author’s short name and the year of the prediction. Alternating background bands are the five era strata, labelled E1 (1800-1870) through E5 (1995 to present). The corpus is purposively selected to exercise the schema and is not representative of the discourse, so the density of points across eras carries no information about the frequency of predictions.

Reassurance is not a modern reaction to modern alarm. Andrew Ure was defending the factory system in 1835 (Ure 1835), and the United States Commissioner of Labor was describing displacement as “only a temporary displacement of labor” in 1886 (Wright 1886). Six of the twenty-three records are reassurances and they are spread across all five eras, which matters for the design rather than merely for the description: any calibration analysis requires the claims that turned out to be right as well as the claims that turned out to be wrong, and a corpus restricted to alarms could only ever produce a list of failures. This is why the scope statement admits reassuring predictions explicitly, and why panic_valence is a field rather than an inclusion criterion.

6.2 Specificity and the Case for Tiering

Figure 3 shows how the four specificity tiers are distributed across eras.

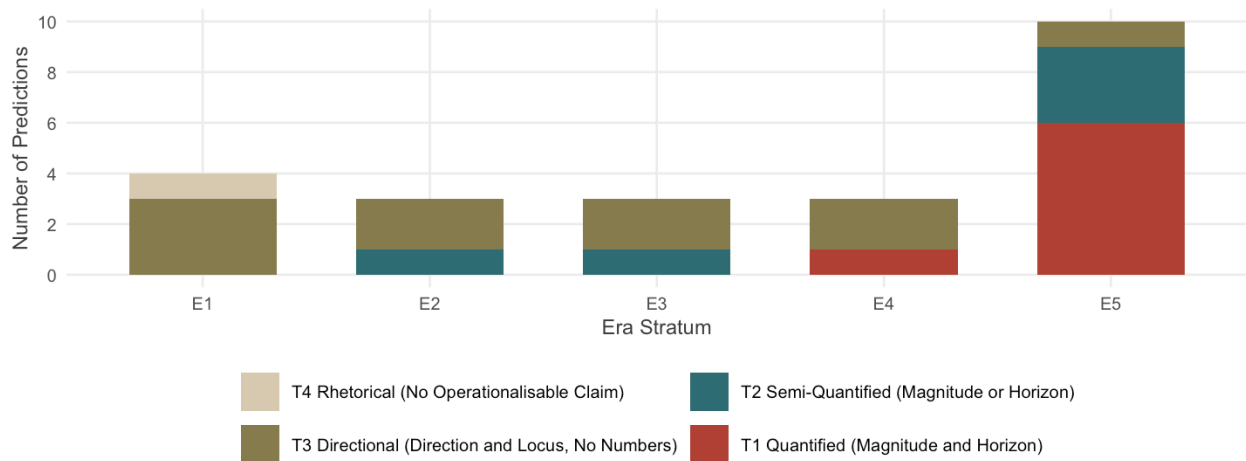


Figure 3: Specificity tier by era stratum, 23 pilot predictions. Notes: The specificity tier records how operationalisable a claim is. T1 (quantified) carries both an explicit magnitude and an explicit horizon, as in a claim that a stated percentage of jobs will be lost by a stated year. T2 (semi-quantified) carries one of the two but not both, as in the Frey and Osborne figure of 47 per cent with a horizon of ‘perhaps a decade or two’. T3 (directional) states a clear direction and a clear locus without numbers, as in a claim that the power loom will throw the hand-weavers out of work. T4 (rhetorical) offers vague alarm or reassurance with no operationalisable claim. Era strata are E1 (1800-1870) through E5 (1995 to present). The corpus is purposively selected to include all four tiers, so the totals within each era should not be read as tier frequencies in the discourse; the era pattern in which tiers appear at all is the object of interest.

Putting a number on the claim is a recent habit. Every T1 record in the pilot dates from 1960 or later, and everything before 1900 is directional or rhetorical. The sample is small and the exact proportions should not be leaned on, but the direction is unmistakable and it justifies a design choice that would otherwise look like indulgence. Tiering claims rather than excluding the unquantified ones is what keeps the first century and a half of the record in the corpus; a design that admitted only quantified forecasts would delete E1 and E2 entirely and would then report, quite spuriously, that technological-unemployment predictions began in the 1960s. The cost of tiering is that verdicts are not comparable across tiers, which is why T3 claims are capped at mostly correct and T4 claims are unfalsifiable by construction.

! To discuss with Ben

I don’t think I’ve searched hard enough for T1 claims in the early eras for the pilot. Every quantified record in the pilot is from 1960 or later, and if that’s real it’s a lovely finding. But it might just be that I went looking in the obvious places and the obvious places are all modern.

Second, the tiers. Are four the right number, and is the T1/T2 line in the right place? At the moment it turns entirely on whether a horizon is explicit, which puts Frey and Osborne in T2 on the strength of “perhaps a decade or two”. That feels harsh for a paper carrying a number as specific as 47 per cent.

Figure 4 plots the subset of claims that carry both an explicit magnitude and an explicit horizon, which is the subset on which the calibration hypotheses can be estimated.

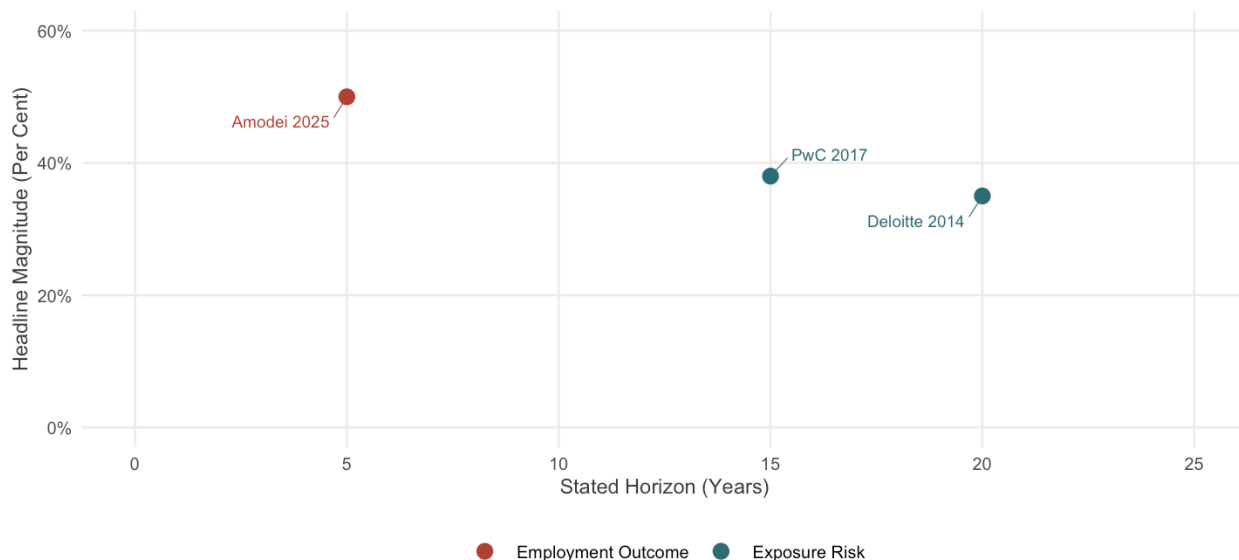


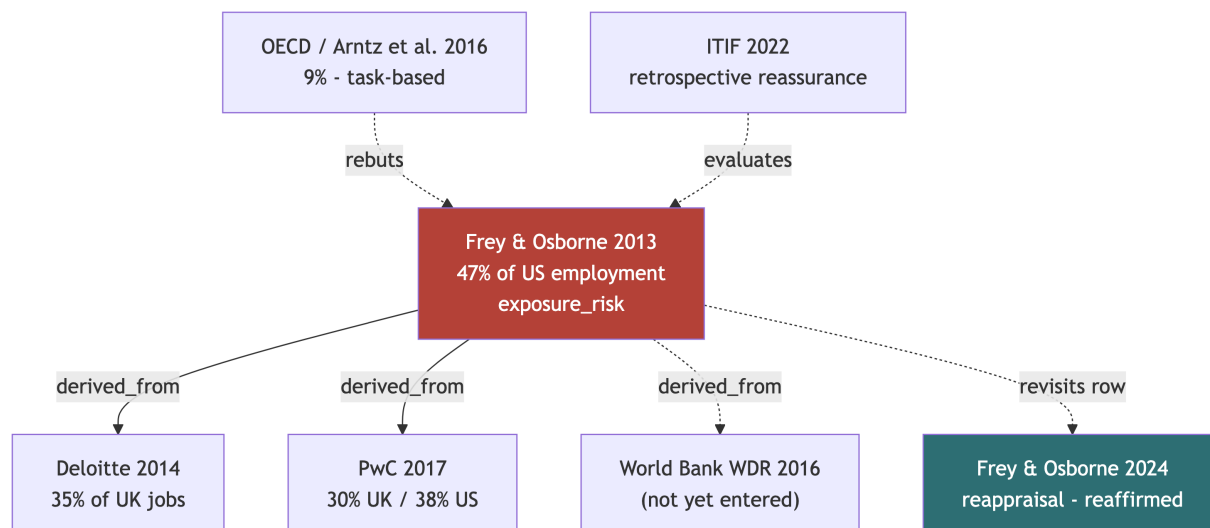
Figure 4: Stated horizon against headline magnitude for the pilot predictions carrying both, 1960-2025. Notes: The horizontal axis is the length of the stated horizon in years, computed as the difference between the horizon end year and the horizon start year. The vertical axis is the headline magnitude, expressed as a percentage of jobs or of employment, and is the single number the author or institution promoted publicly, which for a range is often the more alarming endpoint. Marker colour is the claim type as defined in the notes to Figure 2. Only claims carrying both an explicit magnitude and an explicit horizon can appear, which is why the panel is sparse; the predictions excluded and the reasons for their exclusion are enumerated in the text. Claims measured in absolute counts of jobs rather than shares fall outside this scale entirely.

Figure 4 contains three points out of twenty-three predictions, and that yield is the most useful thing it reports. The predictions that fall out do so for reasons that are each a design question in disguise. The OECD estimate of 9 per cent carries a magnitude but offers only “over the next decades” as a horizon (Arntz and Organisation for Economic Co-operation and Development 2016). Simon’s 1960 claim has a horizon but no number, since “any work that a man can do” is a magnitude of a sort but not one that can be placed on an axis. Frey and Osborne have the number, but their horizon of “perhaps a decade or two” is one the extraction protocol quite properly refuses to convert into a year. Goldman Sachs states 300 million jobs, an absolute count rather than a share, and therefore lies off this scale entirely (Goldman Sachs Global Investment Research 2023). Hinton gave five years and no percentage (Hinton 2016).

The consequence for the analysis plan is worth stating plainly. The hypothesis that accuracy falls as the horizon lengthens, and the hypothesis that realised magnitudes are systematically below claimed magnitudes, are both estimated on this subset. With three points neither is estimable, and the pilot cannot pretend otherwise. What the figure establishes is the shape of the test and the rate at which claims survive to take it, and that rate is the input the precision analysis needs before Registration 2 can state minimum-N gates for each contrast.

6.3 Citation Lineage and the Independence Problem

Several of the most-quoted automation figures are not independent estimates of anything. Frey and Osborne’s 47 per cent was crosswalked onto United Kingdom occupational data by Deloitte in 2014 (Deloitte LLP 2014) and extended by PwC in 2017 (PricewaterhouseCoopers 2017), and the World Bank’s 2016 *World Development Report* (World Bank 2016) and the International Labour Organization’s ASEAN study (Chang and Huynh 2016) both lean on the same underlying methodology. Counting these as four corroborating forecasts would overstate the apparent consensus considerably, and would break the independence assumption underneath every accuracy statistic the project might compute.



The lineage field answers this, and the registered robustness suite re-runs the headline analyses with all derivative predictions dropped, so that one upstream methodology is not counted five times. The dashed edges in the diagram are deliberately not lineage. The OECD task-based re-estimate that puts the figure at 9 per cent rather than 47 is a rebuttal rather than a derivative (Arntz and Organisation for Economic Co-operation and Development 2016), and the ITIF piece of 2022 evaluates the claim rather than restating it (Atkinson 2022). Only the solid arrows set the lineage field.

The Deloitte case illustrates why the field is needed at the level of the record rather than the level of the citation. The report itself never printed an absolute number, only the share of 35 per cent. On the day of publication one newspaper multiplied that share by the Office for National Statistics employment count and reported ten million jobs at risk, while another outlet performed the same arithmetic and arrived at eleven million. Neither figure is Deloitte's and neither is Frey and Osborne's. The corpus records the 35 per cent, and the the derived figures are the sort of thing the secondary-report flag exists to catch.

The same family of records supplies the pilot's illustration of the fourth research question. In February 2024 Frey and Osborne returned to their own number in the *Brown Journal of World Affairs* (Frey and Osborne 2024), restating it as a claim that 47 per cent of jobs are exposed to recent advances in machine learning and robotics, and concluding that the bottlenecks identified in the 2013 paper continue to constrain automation for the foreseeable future. That is a revisit with the stance recorded as reaffirmed. It is also a warning about what is being measured, because an author's own framing of a claim can drift over a decade, and "exposed" is doing different work in 2024 from the work it did in 2013. The revisits table records what the authors said later without touching what they said at the time, and whether the drift is a clarification or a softening is exactly the judgement the fourth research question exists to make.

7 Trial Evaluations

Six predictions were graded under the draft rubric in order to test the bands and the decision rules before Registration 2 freezes them. Table 6 reports the verdicts, the resolution year used, and the number of evidence rows cited in support of each. These are illustrative and are not registered verdicts.

Table 6: Six trial evaluations under the draft rubric. Notes: The resolution year is the year at which the outcome was measured, snapped to the nearest year for which the relevant series is available; a dash indicates that no resolution year applies because the horizon has not elapsed. The evidence-row count is the number of outcome-evidence records cited in support of the verdict; the validation pipeline rejects any graded verdict with none, and rejects a verdict at either extreme of the scale whose evidence is not official statistics, a census series, or an academic study. These verdicts were graded under a rubric that has not been registered and must not be cited.

Prediction	Verdict	Resolved	Evidence rows
Simon 1960	clearly wrong	1980	2
Rifkin 1995	mostly wrong	2020	2
Compton 1938	mostly correct	1970	2
Frey & Osborne 2013	mixed	2019	2
Hinton 2016	mostly wrong	2021	1
Amodei 2025	too early to tell	-	0

Simon’s 1960 claim (Simon 1960) is graded clearly wrong. By 1980, deployed artificial intelligence consisted of a small number of narrow rule-based expert systems: DENDRAL interpreting mass spectra, MYCIN selecting antibiotics without ever being used on a patient, and XCON configuring minicomputer orders at Digital. Nothing performed general-purpose robotics, open-domain language, or commonsense reasoning. The direction of the claim is contradicted rather than merely over-scaled, which is what separates the lowest band from the one above it. The verdict is assigned on capability alone, which keeps the evaluation clear of the argument among historians of the field about the first artificial-intelligence winter.

Rifkin’s 1995 claim (Rifkin 1995) is graded mostly wrong. United States nonfarm employment rose 21 per cent by 2020 and 35 per cent by 2025 on the Bureau of Labor Statistics establishment series, and the employment-to-population ratio fell only 2.1 points between 1995 and 2019. Manufacturing employment, however, fell 30 per cent over the same window. The aggregate claim fails while its sectoral version survives, and that combination is precisely why the verdict is not clearly wrong.

Compton’s 1938 claim (Compton 1938) is graded mostly correct. United States employment rose from 47.5 million in 1940 to 78.6 million in 1970 while unemployment fell from 14.6 per cent to 4.9 per cent. The verdict is capped at mostly correct because the claim is directional and because “in the long run” had to be assigned a horizon by the default rule. One caution belongs with the numbers: the 1940 unemployment rate quoted here is Lebergott’s (Lebergott 1964), and Darby’s treatment, which counts relief workers as employed, gives 9.5 per cent instead (Darby 1976). A five-point spread on the baseline year is large relative to the change being asserted, and the evidence row names the series used.

Frey and Osborne’s 2013 claim (Frey and Osborne 2013) is graded mixed, on the exposure track. The differential test passes: across 21 OECD countries between 2012 and 2019, employment in the riskiest half of occupations grew 6.1 per cent against 17.8 per cent for the least risky half (Georgieff and Milanez 2021). No country saw net job destruction, however, and Coelli and Borland (2019) find that the risk scores add no forecasting value over ordinary routine-task-intensity measures. The ranking held and the implied scale did not, which is what the middle band is for.

Hinton’s 2016 claim about radiologists (Hinton 2016) is graded mostly wrong. By 2021 deep learning had not displaced radiologists and the specialty was in its largest recorded shortage. The direction of travel was not contradicted, and Hinton has since conceded the timing while maintaining the direction, so the verdict is not clearly wrong. The evidence row supporting it is the weakest in the set, being a commentary by a radiology resident in a general-interest magazine, and it would not be accepted in support of a verdict at either extreme of the scale. It should be replaced by an occupational employment series for radiologists before this record leaves draft status.

Amodei’s 2025 claim (Amodei 2025) receives the terminal code for too early to tell, since its horizon runs to 2030. It enters the dataset as right-censored, and a later evaluation cohort will append a graded verdict as a new row rather than overwriting this one. The design treats the dataset as a living instrument for exactly this reason: today’s generative-artificial-intelligence predictions become evaluable in later cohorts without rewriting the record of what was known in 2026.

Grading six predictions taught two lessons, and both concern the rubric rather than the predictions. The first is that both mostly-wrong verdicts turn on locus. In each case an aggregate claim fails while its sectoral version survives. The rubric does contain a band for a claim that is right in direction and wrong in locus, but the pilot suggests that the harder problem lies upstream of the bands, in deciding which claim was being made when the author was vague about scope. That decision belongs in the codebook as an explicit rule rather than in the evaluator’s head, and it is the clearest case in the pilot of a researcher degree of freedom that registration must close.

The second lesson is that two of the six verdicts required a defaulted horizon. Compton’s “in the long run” and Rifkin’s “in the years ahead” are not dates, and the rubric converts them into dates by convention. The registered robustness suite must therefore be able to drop every defaulted-horizon verdict and report what survives. Six trial cases produced two such verdicts, and at corpus scale this will not be a footnote.

! To discuss with Ben — do these six verdicts feel right?

Have a look at the six above and tell me whether you’d have graded any of them differently.

The two I’m least comfortable with are the mostly-wrong pair, because both relate to the same thing: the aggregate claim fails and the sectoral version survives. Rifkin is wrong about total employment and right about manufacturing. I don’t think that’s a band problem. I think it’s a codebook problem - we need a rule for deciding which claim was being made when the author was vague about scope, and I’d rather write that rule with you than guess at it and have it come back at us in review.

The other one is defaulted horizons. Two of six needed one. If that rate holds we’ll be defaulting on the horizon for something like a third of the corpus, which is a lot of researcher discretion...

8 The Extraction Pipeline in Practice

The stage that makes a full corpus feasible rather than merely desirable is language-model extraction, and the pilot therefore ran the pipeline on a passage the model had not previously seen. The model was `gemini-3.7-flash` at temperature zero, constrained to a Pydantic schema mirroring the predictions table, with the system prompt read at run time from the registered prompt file. Pinning the exact model matters for the reason given in Section 3.

The input was a passage from Frey’s *The Technology Trap* (2019, 320):

Figure 17 plots the exposure of major occupational categories to automation by their employment share. Office and administrative support, production, transportation and logistics, food preparation, and retail jobs loom large in terms of both their exposure to automation and the percentage of Americans they support. Overall, our algorithm predicted that 47 percent of American jobs are susceptible to automation, meaning that they are potentially automatable from a technological point of view, given the latest computer-controlled equipment and sufficient relevant data for the algorithm to draw upon. What most of these jobs have in common is that they are low-income jobs that do not require high levels of education (figure 18).

The returned record is reproduced in part below.

<code>claim_type</code>	<code>exposure_risk</code>
<code>level</code>	<code>country_region</code>
<code>geography</code>	USA
<code>direction</code>	displacement
<code>magnitude_type</code>	percent_of_jobs
<code>magnitude_value</code>	47
<code>specificity_tier</code>	T2_semi_quantified
<code>horizon_type</code>	none

mechanism_type	capability_parity
panic_valence	neutral_forecast
is_secondary_report	FALSE

The record is well formed, and the claim is typed as an exposure risk rather than an employment outcome, which is the distinction on which the entire evaluation protocol depends. The model also flagged its own uncertainty without being prompted to do so, noting in the coder-notes field that the author is restating the well-known 2013 finding, that the horizon of ten to twenty years from the original paper does not appear in the excerpt, and that soft hyphens introduced by optical character recognition have been preserved in the verbatim quotation. That note is the pipeline working as intended: it identifies the boundary case a human must resolve, since a passage in which an author restates his own earlier prediction belongs in the revisits table rather than entering as a new prediction, and it records a transcription hazard that must not be silently cleaned up.

Three fields nevertheless required correction at verification, and each failure is of a different kind. The first is an invented vocabulary term: the model proposed a technology identifier that does not exist, where the controlled vocabulary holds artificial intelligence and machine learning as separate entries. Schema-constrained generation prevents the model from inventing structure but does not prevent it from inventing values in a free-text field. The second is a contestable field rather than an error: the model recorded the passage as not a secondary report, and an author restating his own published finding is arguably exactly that, so the codebook needs a rule that settles it. The third is a date that does not appear in the passage at all: the model set the prediction date to the year of the book rather than the year of the prediction, which is 2013.

A separate smoke test made the case for full verification more sharply than expected. Given Simon's twenty-year claim with no metadata attached, the model produced a clean record and volunteered that the passage was a famous quotation of 1965. The quotation is genuine and the date is the standard misattribution; the claim comes from *The New Science of Management Decision* of 1960, reprinted in 1965 (Simon 1960). Correcting exactly this class of error is one of the reasons the seed list exists. A model that is right about the substance and confidently wrong about the provenance is the whole argument for verifying every field rather than spot-checking a sample, since a sample-based check would have found the record clean.

The scope rule was applied correctly and unprompted in the negative case. Given a retrospective passage from the same book, concerning computer adoption in the 1980s, the model declined to extract anything and recorded that the passage contains retrospective historical analysis of past automation anxiety and the productivity paradox but no forward-looking prediction about employment. Retrospective statements are out of scope and only forward-looking claims are in, and this is the behaviour the detection-recall check will measure at scale. Observing it without prompting is encouraging, though a single instance establishes very little.

9 Discussion

The pilot establishes three things about the instrument and surfaces one substantive finding. The schema holds real historical material, including a pseudonymous threatening letter, a maiden speech, a recorded conference talk, and a consultancy report, without requiring any of them to be distorted into a shape they do not have. The pipeline runs end to end, from source text through a schema-constrained draft record to a graded verdict backed by cited evidence, with every field-level correction logged. The rubric produces defensible verdicts on six predictions, and the two places where it strains are now identified. The substantive finding is that the boundary between the second and third era strata does not rest on anything, and that the evidence points to 1928.

A second finding is more tentative and concerns the second stratum as a whole. The period from 1870 to 1920 is the hardest to populate, and the obvious reading of a thin stratum is that the queries are failing. Woirol (1996, 20) reports, however, that after the middle of the nineteenth century the employment effects of technological change ceased to be treated as a relevant problem by the economics profession, even while congressional and popular concern continued (Atack, Margo, and Rhode 2019). If that reading is right, the thinness of E2 is a finding about the professional discourse rather than a hole in the method, and the limitations section should report it as such rather than apologise for it. The per-era retrieval validation described in Section 4, in which a fixed random sample of archive pages per stratum is hand-read and the query miss rate estimated, is what will distinguish the two explanations. Until that validation is run, the question stays open.

The pilot also measured a cost that had not been anticipated at the right order of magnitude. Every source URL in the corpus required a Wayback snapshot in order to satisfy the validation rule that a live URL must be accompanied by an archive permalink. Eighteen of nineteen already had one and the last had to be triggered by hand. At corpus scale this is a pipeline stage rather than a chore, and it should be built as one. The justification for the rule is visible within the pilot itself: one live URL in the corpus, Deloitte's own copy of its own 2014 report, is already an infinite redirect loop, and the document is recoverable only because a byte-identical copy happens to sit on a server in Oxford. Link rot at this rate over a two-century corpus would make a substantial share of the dataset unverifiable within a decade of publication.

Several boundary cases were resolved during the pilot and are recorded because they will recur. Mill's 1848 discussion (Mill 1848) is excluded as retrospective rather than predictive, notwithstanding that it is quoted as a prediction constantly. The ITIF piece of 2022 is a retrospective reassurance, which is to say a claim about the past that functions as a forecast about the future, and it enters as a prediction on those grounds. Goldman Sachs is now coded with an author type of financial institution but a source type of consultancy report, because one vocabulary was split during the pilot and the other was not; the asymmetry is discussed in Section 10.

A final observation concerns quotations, and it is the sort of thing that a dataset of predictions must get right because a dataset of predictions is a dataset of quotations. Four of the most-circulated lines in this literature should never enter the corpus in the form in which they circulate. The remark about the factory of the future being staffed by a man and a dog is an anonymous joke from a trade magazine of 1978 that Warren Bennis, to whom it is attributed, disclaimed. The exchange in which Walter Reuther asks a Ford executive how robots will buy cars is folklore with no documented source. Keynes's passage of 1930 (Keynes 1930) is genuine, but it describes technological unemployment explicitly as "only a temporary phase of maladjustment", which is close to the opposite of the way it is usually deployed. And Simon's twenty-year claim is of 1960, not 1965. The apocryphal-quotation hazard list is wired into the extraction protocol for this reason.

References

The list below covers the scholarly literature cited above and, in addition, every source in the pilot corpus, including the records that appear only in Tables 4 and 5. Corpus entries are generated from `data/processed/sources.csv` at render time, so a reference cannot drift from the record it describes.

Amodei, Dario. 2025. "Behind the Curtain: A White-Collar Bloodbath." *Axios*. <https://www.axios.com/2025/05/28/ai-jobs-white-collar-unemployment-anthropic>.

Armstrong, Stuart, Kaj Sotala, and Seán S. Ó hÉigearthaigh. 2014. "The Errors, Insights and Lessons of Famous AI Predictions – and What They Mean for the Future." *Journal of Experimental and Theoretical Artificial Intelligence* 26 (3): 317–42. <https://doi.org/10.1080/0952813X.2014.895105>.

Arntz, Melanie, and Organisation for Economic Co-operation and Development. 2016. "The Risk of Automation for Jobs in OECD Countries: A Comparative Analysis." OECD Social, Employment; Migration Working Papers No. 189. <https://doi.org/10.1787/5jlz9h56dvq7-en>.

Atack, Jeremy, Robert A. Margo, and Paul W. Rhode. 2019. "'Automation' of Manufacturing in the Late Nineteenth Century: The Hand and Machine Labor Study." *Journal of Economic Perspectives* 33 (2): 51–70. <https://doi.org/10.1257/jep.33.2.51>.

Atkinson, Robert D. 2022. "Oops, the Predicted 47 Percent of Job Loss from AI Didn't Happen." ITIF Blog. <https://itif.org/publications/2022/09/30/oops-the-predicted-47-percent-of-job-loss-from-ai-didnt-happen/>.

Autor, David H. 2015. "Why Are There Still so Many Jobs? The History and Future of Workplace Automation." *Journal of Economic Perspectives* 29(3). <https://doi.org/10.1257/jep.29.3.3>.

Bix, Amy Sue. 2000. *Inventing Ourselves Out of Jobs? America's Debate over Technological Unemployment, 1929–1981*. Baltimore, MD: Johns Hopkins University Press.

Byron, George Gordon. 1812. "Speech on the Frame Work Bill, House of Lords, 27 February 1812." Hansard, HL Deb vol 21 cc964-79. <https://historyofinformation.com/detail.php?id=4654>.

Chang, Jae-Hee, and Phu Huynh. 2016. "ASEAN in Transformation: The Future of Jobs at Risk of Automation." Geneva: International Labour Office, Bureau for Employers' Activities, Working Paper No. 9.

- Coelli, Michael, and Jeff Borland. 2019. "Behind the Headline Number: Why Not to Rely on Frey and Osborne's Predictions of Potential Job Loss from Automation." Melbourne: Melbourne Institute Working Paper No. 10/19.
- Committee on Recent Economic Changes of the President's Conference on Unemployment. 1929. *Recent Economic Changes in the United States*. New York: National Bureau of Economic Research.
- Compton, Karl T. 1938. "The Bogey of Technological Unemployment." *Technology Review*. <https://www.technologyreview.com/2024/01/27/1087041/technological-unemployment-elon-musk-jobs-ai/>.
- Darby, Michael R. 1976. "Three-and-a-Half Million U.S. Employees Have Been Misled: Or, an Explanation of Unemployment, 1934–1941." *Journal of Political Economy* 84 (1): 1–16. <https://doi.org/10.1086/260406>.
- Deloitte LLP. 2014. "London Futures - Agiletown: The Relentless March of Technology and London's Response." Deloitte London Futures. <https://www.robots.ox.ac.uk/~mosb/public/pdf/299/uk-london-futures-agiletown-nov-14.pdf>.
- Diebold, John. 1952. *Automation: The Advent of the Automatic Factory*. New York: D. Van Nostrand.
- Frey, Carl Benedikt. 2019. *The Technology Trap: Capital, Labor, and Power in the Age of Automation*. Princeton, NJ: Princeton University Press.
- Frey, Carl Benedikt, and Michael A. Osborne. 2013. "The Future of Employment: How Susceptible Are Jobs to Computerisation?" Oxford Martin School Working Paper. https://oms-www.files.svdcn.com/production/downloads/academic/The_Future_of_Employment.pdf.
- . 2024. "Generative AI and the Future of Work: A Reappraisal." *Brown Journal of World Affairs* 30(1). <https://ora.ox.ac.uk/objects/uuid:f52030f5-23eb-4481-a7f1-8006685edbae>.
- Georgieff, Alexandre, and Anna Milanez. 2021. "What Happened to Jobs at High Risk of Automation?" Paris: OECD Social, Employment; Migration Working Papers No. 255. <https://doi.org/10.1787/10bc97f4-en>.
- Goldman Sachs Global Investment Research. 2023. "The Potentially Large Effects of Artificial Intelligence on Economic Growth." Goldman Sachs Global Economics Analyst. <https://www.goldmansachs.com/insights/articles/generative-ai-could-raise-global-gdp-by-7-percent>.
- Gompers, Samuel. 1894. "Letter on Labor in Industrial Society, to Judge Peter Grosscup." <https://www.let.rug.nl/usa/documents/1876-1900/samuel-gompers-letter-on-labor-in-industrial-society-to-judge-peter-grosscup-september-1894.php>.
- Gordon, Robert J. 2012. "Is U.S. Economic Growth over? Faltering Innovation Confronts the Six Headwinds." Cambridge, MA: National Bureau of Economic Research Working Paper 18315. <https://doi.org/10.3386/w18315>.
- . 2016. *The Rise and Fall of American Growth: The U.S. Standard of Living Since the Civil War*. Princeton, NJ: Princeton University Press.
- Herrington, Gaya. 2021. "Update to Limits to Growth: Comparing the World3 Model with Empirical Data." *Journal of Industrial Ecology* 25 (3): 614–26. <https://doi.org/10.1111/jiec.13084>.
- Hinton, Geoffrey. 2016. "Remarks at the Machine Learning and the Market for Intelligence Conference, Toronto." <https://www.youtube.com/watch?v=2HMPRXstSvQ>.
- Hounshell, David A. 1995. "Automation, Transfer Machinery, and Mass Production in the U.S. Automobile Industry in the Post-World War II Era." *Business and Economic History* 24 (1): 59–71.
- Katselidis, Ioannis, Angelos Vouldis, and Panayotis G. Michaelides. 2011. "The Political Economy of Technological Unemployment." *International Journal of Social Economics* 38 (6): 538–50.
- Keilman, Nico. 2001. "Data Quality and Accuracy of United Nations Population Projections, 1950–95." *Population Studies* 55 (2): 149–64. <https://doi.org/10.1080/00324720127700>.
- Keynes, John Maynard. 1930. "Economic Possibilities for Our Grandchildren." *The Nation and Athenaeum*. <https://www.marxists.org/reference/subject/economics/keynes/1930/our-grandchildren.htm>.
- Landes, David S. 1969. *The Unbound Prometheus: Technological Change and Industrial Development in Western Europe from 1750 to the Present*. Cambridge: Cambridge University Press.

- Lebergott, Stanley. 1964. *Manpower in Economic Growth: The American Record Since 1800*. New York: McGraw-Hill.
- Leontief, Wassily. 1983. "Technological Advance, Economic Growth, and the Distribution of Income." *Population and Development Review* 9(3). <https://doi.org/10.2307/1973315>.
- Loungani, Prakash. 2001. "How Accurate Are Private Sector Forecasts? Cross-Country Evidence from Consensus Forecasts of Output Growth." *International Journal of Forecasting* 17 (3): 419–32. [https://doi.org/10.1016/S0169-2070\(01\)00098-3](https://doi.org/10.1016/S0169-2070(01)00098-3).
- Mill, John Stuart. 1848. *Principles of Political Economy, with Some of Their Applications to Social Philosophy*. London: John W. Parker.
- Mokyr, Joel. 1999. "The Second Industrial Revolution, 1870–1914." In *Storia Dell'economia Mondiale*, edited by Valerio Castronovo, 219–45. Rome: Laterza.
- Mokyr, Joel, Chris Vickers, and Nicolas L. Ziebarth. 2015. "The History of Technological Anxiety and the Future of Economic Growth: Is This Time Different?" *Journal of Economic Perspectives* 29 (3): 31–50. <https://doi.org/10.1257/jep.29.3.31>.
- "Ned Ludd" / Army of Redressers. 1812. "Letter to Mr Smith, Shearing Frame Holder at Hill End, Yorkshire." The Writings of the Luddites (Binfield, ed., 2004). <http://ludditebicentenary.blogspot.com/2012/03/9th-march-1812-letter-from-ned-ludd.html>.
- Oliner, Stephen D., and Daniel E. Sichel. 2000. "The Resurgence of Growth in the Late 1990s: Is Information Technology the Story?" *Journal of Economic Perspectives* 14 (4): 3–22. <https://doi.org/10.1257/jep.14.4.3>.
- PricewaterhouseCoopers. 2017. "Will Robots Steal Our Jobs? The Potential Impact of Automation on the UK and Other Major Economies." UK Economic Outlook, March 2017. <https://www.pwc.co.uk/economic-services/assets/international-impact-of-automation-feb-2018.pdf>.
- Reuther, Walter. 1955. "Testimony Before the Subcommittee on Economic Stabilization, Joint Committee on the Economic Report: Automation and Technological Change." [https://www.jec.senate.gov/reports/84th%20Congress/Automation%20and%20Technological%20Change%20-%20Hearings%20\(75\).pdf](https://www.jec.senate.gov/reports/84th%20Congress/Automation%20and%20Technological%20Change%20-%20Hearings%20(75).pdf).
- Ricardo, David. 1821. *On Machinery (Chapter 31)*. On the Principles of Political Economy and Taxation, 3rd Edition. <https://www.marxists.org/reference/subject/economics/ricardo/tax/ch31.htm>.
- Rifkin, Jeremy. 1995. *The End of Work: The Decline of the Global Labor Force and the Dawn of the Post-Market Era*. <http://pinguet.free.fr/rifkin1995.pdf>.
- Simon, Herbert A. 1960. *The New Science of Management Decision*.
- Steuart, James. 1767. *An Inquiry into the Principles of Political Oeconomy*. London: A. Millar; T. Cadell.
- Tetlock, Philip E. 2005. *Expert Political Judgment: How Good Is It? How Can We Know?* Princeton, NJ: Princeton University Press.
- "The Price of Industrial Progress." 1928. *The New Republic*.
- Timmermann, Allan. 2007. "An Evaluation of the World Economic Outlook Forecasts." *IMF Staff Papers* 54 (1): 1–33. <https://doi.org/10.1057/palgrave.imfsp.9450007>.
- Turner, Graham M. 2008. "A Comparison of The Limits to Growth with Thirty Years of Reality." *Global Environmental Change* 18 (3): 397–411. <https://doi.org/10.1016/j.gloenvcha.2008.05.001>.
- United States Congress, Joint Committee on the Economic Report. 1955. "Automation and Technological Change: Hearings Before the Subcommittee on Economic Stabilization." 84th Congress, 1st Session, 14–28 October 1955. Government Printing Office, Washington, DC.
- Ure, Andrew. 1835. *The Philosophy of Manufactures*. https://wikieducator.org/images/c/c9/Modern_History_Sourcebook_URE.pdf.
- Wells, David A. 1889. *Recent Economic Changes*. https://en.wikisource.org/wiki/Popular_Science_Monthly/Volume_35/September_1889/Recent_Economic_Changes.

Wiener, Norbert. 1949. "Letter to Walter Reuther, 13 August 1949." <https://libcom.org/history/father-cybernetics-norbert-wieners-letter-uaw-president-walter-reuther>.

Woirol, Gregory R. 1996. *The Technological Unemployment and Structural Unemployment Debates*. Westport, CT: Greenwood Press.

World Bank. 2016. *World Development Report 2016: Digital Dividends*. Washington, DC: World Bank. <https://doi.org/10.1596/978-1-4648-0671-1>.

Wright, Carroll D. 1886. "First Annual Report of the Commissioner of Labor: Industrial Depressions."

Repository: github.com/j-jayes/tech-panic-tracker.